

When Women Get a Chance: Direct and Spillover Effects of Women's Access to Higher Education in Pakistan

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Abstract

Female enrollment in higher education in developing countries remains low due to cultural norms restricting mobility and concerns about safety. This paper examines whether expanding local educational opportunities can help mitigate these barriers. I study a novel policy in Pakistan's Khyber Pakhtunkhwa (KP) province that opened previously male-only public colleges to female students by introducing a four-year BS program. Combining administrative data on the rollout of the BS program with the Pakistan Social and Living Standards Measurement (PSLM) survey, I use a difference-in-differences design to show that increased geographic access to higher education raises female schooling and the likelihood of completing a four-year degree. Consistent with dynamic complementarities in human capital investment, I also document positive spillover effects for younger girls at the primary, secondary, and post-secondary levels. Finally, I find suggestive evidence of improvements in marriage-market outcomes for post-college-aged women.

Keywords: Higher Education; Colleges; Coeducation; Public Infrastructure

JEL Codes: H52, I21, I23, I25, O15.

1 Introduction

Over the past two decades, the developing world has made exemplary progress in expanding primary and secondary schooling and increasing female enrollment at the pre-college level (UNESCO 2020). Yet, enrollment rates of females at tertiary education remain low (Sekhri, Hossain, and Khosla 2022). In 2021, the female gross enrollment ratio in tertiary education was just 7.4 percent in low-income countries,¹ compared to 87.4 percent in OECD countries (UNESCO 2026). A growing body of research shows that concerns about safety and restrictive gender norms imposed by religion and culture substantially constrain women’s mobility, thereby reducing female enrollment in higher education in low-income countries (Borker et al. 2021; Jayachandran 2015; Sekhri, Hossain, and Khosla 2022). Expanding the local supply of higher education institutions can help ease these constraints (Ahmed 2024; Zhang and R. Assaad 2024). However, unlike investments in primary and secondary schooling, public spending on tertiary education remains politically contentious for two main reasons. First, unlike spending on primary or secondary schooling, public expenditure on higher education is often perceived as regressive, which raises concerns about its distributional consequences (Birdsall 1996; Diris and Ooghe 2018; Garcia-Penalosa and Wälde 2000). In addition, labor market returns to higher education for women are also considered limited (Psacharopoulos and Patrinos 2004), which casts doubt on the economic justification for such large investments.

In Pakistan, college education has historically been delivered through gender-segregated public colleges, with coeducation largely confined to public and private universities (Karim 2024; Malik and Mirza 2014). In this paper, I examine a policy reform in Khyber Pakhtunkhwa (KP) that altered this structure. Following the introduction of the four-year BS program under the National Qualification Framework, previously male-only colleges were opened to female students. Because these colleges are geographically dispersed rather than concentrated in urban centers, introducing coeducation at the BS level expanded local access to higher education for women. Unlike the construction of entirely new institutions, this policy leveraged existing infrastructure, representing a comparatively cost-effective way of increasing local access to tertiary education for women.

I exploit the staggered roll-out of the BS program, which expanded access to higher education for women through the introduction of coeducation in existing colleges. Using a two-way fixed effects design and variation in exposure across birth cohorts and districts, I estimate the direct and spillover effects of the program. I find that exposure to higher education

1. A time-series comparison is available [here](#).

increases years of schooling among women who are directly exposed to the program by 12.5 percent and increases the probability of completing higher education by 30 percent. At the same time, exposure to higher education lowers the probability of marriage.² However, I find no labor market returns to exposure to higher education.

Next, I study the effects of exposure to higher education on marriage market outcomes among women who are post-college-going age. The results show that exposure to higher education reduces the educational difference between spouses. I find no effects on assortative mating at the higher education level. However, I find changes in educational matching at other levels, as reflected in patterns of hypergamy, homogamy, and hypogamy. Additionally, I find no effects of exposure to higher education on spousal employment outcomes.

I then examine the spillover effects of higher education expansion on younger cohorts. Using a two-way fixed effects design with district and survey-year variation, I find that greater access to BS colleges increases enrollment among younger female cohorts. In contrast, I find no spillover effects for men, suggesting that the expansion primarily affected women’s educational opportunities at the pre-college level. Taken together, these results show that expanding access to higher education for women through coeducation not only raises educational attainment among directly exposed cohorts but also generates positive spillovers for younger girls. While I find no labor market effects, the results point to other important channels through which access to higher education shapes women’s outcomes.

This paper contributes to two distinct strands of the literature. Recent studies from the developing world show that access to higher education for females increases enrollment in higher education and improves their agency. For instance, Elsayed and Shirshikova (2023) show that access to higher education improves human capital and labor-market outcomes and also yields beneficial effects on marriage outcomes. In another study, Sun and Sun (2023) examine China’s higher-education expansion policy and find that it reduces the gender gap by 8.4%. Similarly, Huang and Lin (2024) study a coeducation reform at Peking University in China that exogenously increased women’s access to university education for the first time. Their paper documents the diffusion of progressive gender norms in counties exposed to the coeducation reform but finds limited spillover effects on female enrollment at primary schools. Furthermore, Sekhri, Hossain, and Khosla (2022) show positive effects of college construction in India on women’s education and document beneficial impacts on women’s labor- and marriage-market outcomes. The existing literature primarily focuses on the expansion of higher education through the construction of new post-secondary institutions, a

2. Unfortunately, the data do not contain information on age at marriage, which would have helped identify whether this reflects a delay in marriage.

distributionally contentious policy choice. In contrast, this paper evaluates a program in which existing male-only colleges introduced coeducation by admitting female applicants, resulting in coeducation within a religiously conservative institutional setting.³

Second, there is a nascent but growing literature documenting the spillover effects of access to higher education on younger cohorts (Jagnani and Khanna 2020; Katzkowicz et al. 2023; Mukhopadhyay and Sahoo 2016; Zhang and R. A. Assaad 2023). Notably, Hertweck and Yasar (2024) study the impacts of university expansion in Germany on younger cohorts and find that the local arrival of a university leads to changes in the educational choices made by younger cohorts, with substantial shifts toward university degrees and away from vocational training. Katzkowicz et al. (2023) show that the expansion of higher-education institutions in Uruguay results in intergenerational mobility by increasing university enrollment among less-privileged groups. In addition, they find increased educational attainment among younger cohorts. Jagnani and Khanna (2020) study the expansion of elite colleges in India and find that the local presence of an elite college increases enrollment among the pre-college-age population.⁴

The rest of the paper is organized as follows. Section 2 describes the institutional background and details of the program. Section 3 presents the data and descriptive statistics. Section 4 outlines the empirical strategy, section 5 discusses the main results, followed by section 6, which presents robustness analyses. Lastly, section 7 concludes.

2 Background

2.1 Education Structure in Pakistan

The national education structure in Pakistan differs from that of the United States and draws inspiration from the British educational system (Saeed 2007). After mandatory pre-school education, schooling is broadly divided into three stages: primary (grades 1–5), middle (grades 6–8), and secondary (grades 9–10). After completing secondary school, students sit for the matriculation examination and move on to intermediate colleges, where they pursue FA/FSc, which is equivalent to 12th grade.

3. Relatedly, at the primary level, Karim (2024) shows that the transition from single-sex to coeducational primary schools led to a 10–19 percent decline in girls’ enrollment in transitioned schools relative to schools that were already coeducational.

4. Although conceptually distinct, studies have also documented positive effects of access to secondary schooling on primary-school enrollment (Ahmed 2024; Mukhopadhyay and Sahoo 2016; Sandholtz 2024).

Following intermediate education, students either enter professional tracks—such as medical or engineering programs—or pursue general higher education. Admission typically depends on subject choice and performance in the 12th grade. Both universities and colleges⁵ offer bachelor’s degrees. Students may enroll in either a BA/BSc (two-year degree) or a BS (four-year degree).

A key distinction of the BS program is its semester system, under which students specialize in a single field of study. Graduates of the BS program are eligible to apply directly to PhD programs abroad.⁶ In the job market, both routes signal the required 16 years of education, and employers typically do not distinguish between them in the hiring process (Dawn 2006).

2.2 The Reform

In 2009, the Higher Education Commission (HEC) of Pakistan introduced the National Qualification Framework to align the country’s education system with international standards. Shortly thereafter, higher education was devolved to the provinces through the 18th Constitutional Amendment, which tasked each province with reforming its higher education system. Since then, progress has been uneven, with significant variation across provinces in the implementation of the 2009 framework.

Khyber Pakhtunkhwa (KP) took the lead by introducing a staggered rollout of the four-year BS program across 320 colleges in the province, while other provinces lagged behind. A key feature of this reform, central to this study, is that it exogenously expanded access to higher education for women. Prior to the reform, higher education was largely restricted to gender-segregated colleges, with universities serving as the primary providers of coeducation. With the introduction of the BS program, however, female students became eligible for admission to formerly male-only colleges on an open-merit basis, determined by matriculation and intermediate examination scores.⁷ Figure 1 presents the timeline of events leading to the expansion of BS colleges to female students, along with the phases of expansion.

These colleges offer degrees equivalent to those of universities, as they are affiliated with

5. Before the reform (discussed in the section 2.2), with few exceptions, most universities in Pakistan offered only a two-year bachelor’s degree (BSc) followed by a master’s degree (MSc). Similarly, most colleges did not offer BS programs.

6. Students who follow the BA/BSc and then MA/MSc route generally need to complete an MPhil (18 years of education) before becoming eligible for PhD admission in Pakistan or abroad.

7. Male colleges are geographically dispersed across districts, whereas female colleges and universities are concentrated in urban areas. As a result, the program unintentionally increased access to higher education for women.

nearby universities and award university-certified degrees.⁸ Moreover, students retain the option to exit after two years with a BA/BSc degree, formally known as an associate degree.

Figure 2 presents the spatial distribution of higher education institutions in the KP province. It is evident from the figure that the colleges (represented by dots) are more spatially dispersed, while universities (represented by stars) are concentrated in urban centers. To address the concern that people may have become aware of the existence of these BS colleges only gradually over time, I use Google search intensity for two terms: “degree college” and “BS.” Given the seasonality in search behavior, I aggregate the Google search data and, following Google Trends format, rescale the aggregated score to 100 for the most popular year–province value.⁹ The results presented in Figure 5 show the popularity of the term “degree college” across three provinces of Pakistan from 2010 to 2014.¹⁰ I also show the evolution of BS colleges in KP province over time. The results indicate that, on average, the term “degree college” is more popular in KP province, where BS colleges expanded most rapidly. Moreover, the relative popularity of the term increases over time as the BS program expands across colleges. Similarly, Figure 4 compares the relative popularity of the terms “BS”, “MSc”, and “MPhil” in KP province. There is a clear upward trend in the popularity of “BS,” while searches for “MSc” and “MPhil” remain largely stagnant over time.¹¹

3 Data and Descriptive Statistics

3.1 Data

I exploit district-level quasi-random variation in the opening of BS colleges in Khyber Pakhtunkhwa (KP), Pakistan, which expanded access to higher education for women. To measure district-level exposure to the BS program and the resulting expansion of higher education opportunities for women, I use administrative records from the KP Higher Education Department, which report the location of colleges and the year in which the BS program

8. The cost of a four-year BS degree at a college is roughly one-quarter that of a university.

9. Google News Initiative (2019) explains that the Google Trend Values are scaled from 0 to 100, where 100 corresponds to the peak search intensity within the selected time period and geography

10. Monthly raw Google search data for the term “degree college” can be found [here](#).

11. Monthly raw Google search data for the terms “BS,” “MSc,” and “MPhil” in KP province can be found [here](#). Alternatively, [this link](#) presents raw Google search data for the terms “BSc,” “MSc,” and “BS.” Notably, “BSc,” which is replaced by “BS,” becomes less popular in later years as the popularity of “BS” rises, further suggesting that local populations were well informed about the transition. Finally, raw Google search comparisons for the term “college merit list” across the three provinces are available at [this link](#). Interestingly, “college merit list” was initially searched at similar rates in Punjab and KP, but became more popular in later years in KP with the expansion of the BS program.

was introduced. Using these data, I construct the number of BS colleges in each district by year.¹² I then create a district-level measure of exposure by dividing the number of BS colleges in a given year by the female population of college-going age, thereby accounting for differences in district size.¹³

To examine the effects of access to BS programs, I use the Pakistan Social and Living Standards Measurement (PSLM) survey, a nationally representative dataset conducted annually by the Pakistan Bureau of Statistics (PBS). The PSLM employs a stratified random sampling design and covers approximately 80,000 households each year. In alternate years, the sample size is reduced to around 30,000 households, with additional modules added to the questionnaire, which is named the Household Integrated Expenditure Surveys (HIES). Both the long and short formats consistently collect information on education, employment, and marriage-related outcomes, ensuring comparability across rounds. To access the direct effects of the program on women’s education, labor, and marriage market outcomes, I use HIES 2019 and PSLM 2020 waves and retroactively assign exposure to colleges to women based on the district of residence and year of birth. For the spillover analysis, I merge all rounds of the cross-sectional PSLM surveys between 2004–05 and 2019–20 to construct a pooled dataset that combines individual- and district-level information.

3.2 Descriptive Statistics

As described in Section 3.1, I construct the data in two distinct ways to study the direct and spillover effects of access to higher education. Tables B1 and B2 report the corresponding descriptive statistics.

Panel A of Table B1 presents summary statistics for the sample used to study the effects of the BS program on women’s educational attainment and enrollment, exploiting the cohort analysis. While constructing the dataset for this analysis, I assign exposure (i.e., district-level BS intensity) based on the year in which a woman turned 19, the expected entry year into a four-year BS program, and her district of residence. I restrict the sample to women in Khyber Pakhtunkhwa province, exclude cohorts with exposure years after 2015, and drop women older than 40. These restrictions ensure that the sample includes cohorts that were potentially exposed to the BS program and are old enough to have completed a four-year degree (i.e., around age 23), as well as older cohorts that are too old to be exposed to the

12. Although the program expanded in a staggered fashion, I am unable to exploit staggered adoption because the outcome data are only available at the district level, which is the lowest administrative unit.

13. Population data are obtained from the district-wise tables of the 7th Population and Housing Census 2023, published by the Pakistan Bureau of Statistics.

BS program rollout.

This sample consists of women aged 23 to 40. On average, women have 3.687 years of schooling, and 5.2 percent have completed 16 years of education.¹⁴ Ten percent of women in the sample are employed in either the formal or informal sector.¹⁵ Panel B presents descriptive statistics for marriage and spousal labor market outcomes. In this sample, 88.6 percent of women are married. The average spousal age gap is 3.38 years, and the average spousal education gap is 3.83 years, consistent with positive educational assortative mating. Spousal labor force participation is 86.2 percent.

Table B2 reports descriptive statistics based on an alternative construction for the spillover effects. Panel A shows that 10.7 percent of women are currently enrolled, with an average of 4.16 years of schooling. Panel B reports enrollment rates by primary-, secondary-, matriculation-, and intermediate- levels, with enrollment at 59.1, 50.7, 35.9, and 22.6 percent, respectively.

Finally, Panel C presents the summary statistics of per capita colleges. In addition, Figure 5 plots the trend in the exposure variable over time. In 2015, there were approximately 0.5 BS colleges per 100,000 girls; however, with the rapid expansion of the program across the province, this number increased to about 2 colleges per 100,000 girls by 2020. Figure B1 illustrates the cross-district variation in college locality, highlighting substantial heterogeneity in access to BS colleges across districts.¹⁶

4 Empirical Strategy

This section outlines the empirical strategy used to study the direct and spillover effects of women’s access to higher education. I begin by describing the empirical strategy used to identify the direct effects of the BS program on college-age women, who were directly exposed to the expansion. I then turn to the spillover effects on younger cohorts, who were indirectly affected through changes in future access to higher education.

14. Average years of schooling in my sample is comparable to that reported in Khan (2021), although the share completing 16 years is lower than the national average of approximately 8 percent reported in UNESCO (2026).

15. This is considerably below the national female labor force participation rate of 24 percent, likely because the sample focuses on a sample from KP Province.

16. Table B9 shows the number of treated and untreated districts by year.

4.1 Direct Effects of Access to Higher Education

To study the direct impact of the BS program on educational and marriage market outcomes among post college-age women who were exposed to the colleges, following Duflo (2001) and Katzkowicz et al. (2023), I estimate the following specification:

$$y_{ibd} = \beta_0 + \beta_1 \text{Intensity}_{bd} + \delta_b + \gamma_d + \eta_t + \varepsilon_{ibd}, \quad (1)$$

where y_{ibd} denotes the outcome of interest for individual i born in cohort b and residing in district d . The variable Intensity_{bd} represents the per capita number of BS colleges in district d at the time when age-cohort b was at the college-going age, capturing exposure to the higher education expansion.¹⁷ The equation includes district fixed effects (γ_d) to account for time-invariant differences across districts, birth-year fixed effects (δ_b) to absorb cohort-specific shocks, and survey-year fixed effects (η_t) to control for aggregate time shocks common across districts. Lastly, ε_{ibd} provides the error term. Standard errors are clustered at the district level.

The coefficient of interest, β_1 , captures the average effect of access to BS colleges on the outcomes of directly exposed women. The main identifying assumption is that, in the absence of the BS program, outcomes in districts with and without BS expansion would have followed similar trends. While the continuous nature of treatment make it hard to formally test the pre-trends, I show descriptive evidence that ever-treated and never-treated districts follow similar pre-program trajectories in outcomes. Another concern is the endogenous rollout of BS colleges in districts with higher demand for female higher education. Unfortunately, I am unable to address this concern directly, but I show in the section – that the results are not driven by any specific district.

4.2 Spillover Effects on Younger Cohorts

Next, to study the spillover effects of the BS program on younger cohorts—those who were pre-college age during the expansion—I estimate the following specification:

17. I assign treatment exposure retroactively using an individual’s age, district of residence, and survey year. For each district–year, I measure exposure as the per capita number of BS colleges in that district. I assign this intensity based on the district of residence and cohort, mapping each individual to the district–year corresponding to age 19. Women aged 28 and above serve as the control group, as these cohorts were no longer of college-going age at the time of the BS expansion.

$$y_{idt} = \beta_0 + \beta_1 \text{Intensity}_{dt} + X'_{dt} \Psi + \mu_d + \eta_t + \gamma_b + \varepsilon_{idt}, \quad (2)$$

where y_{idt} represents the outcome for individual i in district d and survey year t . The variable Intensity_{dt} measures the per capita number of BS colleges in district d in year t , capturing the intensity of exposure to the program at the local level. X_{dt} includes time-varying district-level controls, such as the number of universities, urbanization rates, and employment composition. The model includes district fixed effects (μ_d), year fixed effects (η_t), and birth-year fixed effects (γ_b) to account for unobserved heterogeneity across space, time, and cohorts. Standard errors are clustered at the district level.

The coefficient β_1 captures the average spillover effect of the expansion of BS programs on outcomes for younger cohorts. A causal interpretation of this coefficient relies on several assumptions. First, in the absence of the BS expansion, outcomes for younger cohorts in districts with different levels of exposure would have followed similar trends. Since exposure varies continuously across districts, formally testing pre-trends is challenging. Nevertheless, I provide descriptive evidence showing that districts with higher and lower exposure exhibit similar pre-program trajectories in pre-college outcomes.

A related concern is that the rollout of BS colleges may be endogenous to improvements in local educational conditions. For instance, BS programs may have been introduced in districts where demand for education was already increasing. While I cannot fully rule out this possibility, I show in the robustness section 6 that the expansion of BS programs does not coincide with changes in the number of pre-college public educational institutions and distance to education institutions.

5 Main Results

In this section, I present the estimation results. I start by presenting the results from the cohort-level analysis in section 5.1, which mainly capture the first-order effects of the program, followed by the labor and marriage-market effects on post-college-age females who were exposed to the program. Then, in section 5.2, I present the results where I estimate equation 2 to analyze the spillover effects of access to higher education.

5.1 Evidence from cohort analysis

First order effects

Table 1 documents the first-order effects of access to higher education on women by estimating equation 1. Column 1 presents the effects on years of schooling. Among college-completion-aged women exposed to the program, I find an increase in years of schooling of 0.462 years. Relative to the mean of the dependent variable, this corresponds to an increase of 12.5 percent. Column 2 reports results for the probability of completing 16 years of education (i.e., graduating with a BS degree). The estimates show that access to higher education increases the probability of graduating with 16 years of education by 0.016 percentage points, or 30 percent relative to the baseline. Column 3 presents results for the probability of marriage, which declines by 0.048 percentage points (5.4 percent) as a result of exposure to higher education. Finally, Column 4 reports the effects of exposure to the program on the probability of employment, where I find no statistically significant effects.

These findings are consistent with the existing literature on the effects of access to colleges for women. For instance, Sekhri, Hossain, and Khosla (2022) show that access to higher education increases women’s years of schooling and reduces the probability of marriage. Similarly, Ahmed (2024) report delayed marriage among women exposed to university expansion in Egypt.

Marriage market effects

Next, I examine the effects of access to higher education on the marriage market outcomes of women who are exposed to the program and are past college-going age. Table ?? presents the results. Column 1 reports the effects on the age difference between spouses. Although the point estimate suggests a reduction in the age gap, the effect is not statistically significant. Column 2 shows the effects on educational differences between spouses, and I find that exposure to the program significantly reduces the education gap within couples.

Column 3 presents results for educational assortative mating at the higher education level. I find no evidence that exposure to the program affects assortative mating at higher education. As I show spillover effects of the program in the next section, it is possible that marriage market adjustments occur at levels other than higher education. In this regard, Columns 4–6 report results for hypergamy, homogamy, and hypogamy. I find that exposure to the program reduces educational hypergamy and increases homogamy, with no effect on hypogamy. Finally, Column 7 reports the effects on spousal employment status, where I find

no significant effects.

These results are broadly consistent with the existing literature. Notably, Sekhri, Hossain, and Khosla (2022) and Ahmed (2024) also document changes in educational matching with no significant effects on spousal labor market outcomes in the Indian context. In contrast, Ahmed (2024) find that women exposed to universities are more likely to marry spouses in white-collar occupations.

5.2 Evidence from TWFE analysis

First order effects

Now, to substantiate the results presented in Table 3, I provide the evidence from the TWFE analysis, estimating equation 2. Table 3 shows the first-order effects with a different empirical strategy. Akin to the previous results, Table 3 shows that the exposure to the program results in an increase in current enrollment status and year of schooling among the exposed females. Column 1 shows that the enrollment status of college-going females increases by 0.23 percentage points or 21 percent, accounting for the sample mean. Similarly, years of schooling increased by 0.363 percentage points or 8.73 percent if accounted for the sample mean. These results are in line with what I find in the Table 1.

Spillover effects on younger cohorts

Next, the existing research shows that access to higher education also has spillover effects on pre-college education. This effect works through two different channels. First, the arrival of college locally in environments where there are constraints on female mobility, parents could re-optimize their decision of sending their girls to school. Second, there could be role model effects.

Following the existing literature, I study the effects of exposure to the program on age-specific enrollment for pre-college education of females. Starting with primary schooling, column 1 of the Table 4 shows that the exposure to the program increased the primary enrollment, on average, by 0.012 percentage points (2 percent). Subsequently, column 2 of the Table 4 presents the effects of exposure to the program on the enrollment status of secondary school-aged females. Estimates show that exposure to the program results in an increase in enrollment among this age group by 0.016 percentage points (3.1 percent). Likewise, columns 3 and 4 show the results for the probability of enrollment for matriculation

and intermediate-going-aged females. These results show that exposure to higher education results in an increase in enrollment among these groups by 0.022 (6.1 percent) and 0.017 (7.52 percent). These results are aligned with Jagnani and Khanna (2020), who studies the spillover effects of the opening of elite colleges on the pre-college students in India. Katzkowicz et al. (2023) also shows the spillover effects of access to universities on pre-university educational outcomes.

Dynamic complementarities in educational investments

Finally, to further investigate the spillover effects of access to higher education, I follow the existing literature on dynamic complementarities to assess whether the changes in future educational opportunities result in changes in parental educational investments today. Table 5 analyzes the effects of access to higher education on the educational expenditures and pre-college school choice for females. Column 1 of the Table 5 shows an increase of PKR 432.31 (8.39 percentage increase compared to baseline) in child educational spending.¹⁸ Column 2 reports an increase in enrollment in private schools for exposed females, while enrollment in public schools remains the same. Finally, column 4 reports the probability of being enrolled in a distance learning program.¹⁹ The results show a substantial decline in distance education for females who are exposed to the program. Taken together, Table 5 points to the evidence of dynamic complementarities in parental educational investments.

6 Robustness analysis

In this section, I report several robustness checks. I start by presenting the effects of the program on men. Followed by the results where I present the results for men, I provide results from regressions with an alternate outcome variable and alternate specifications. Then, I present the results from regressions, where I drop one district at a time which is followed by two placebo exercises. Finally, to obviate the concerns about the spillover effects that those are purely a supply-side response rather than changes in preferences of parents, I estimate the effects of higher education expansion for females on the availability of pre-college schools.

18. Educational expenditures regressions rely on the data from HIES Surveys, thus have fewer observations

19. Secondary Schooling Examination Boards and Allama Iqbal Open University offer distance learning programs for post-middle school programs up to the PhD. These programs are popular among rural women (Masaud, Awan, and Ali 2017)

Effects on males

First, I examine the effects of the expansion of higher education for women on men. In theory, extending previously male-only colleges to women could generate negative spillover effects on men’s higher education outcomes. However, we should not expect such spillovers in this context, as the program did not alter men’s future educational opportunities.

To test this, I re-estimate both the first-order and spillover regressions for men. Using the cohort-level sample, I first estimate equation 1 for the male subsample. Table B3 shows that the expansion of higher education for women has no effect on men. I then re-estimate the first-order effects for men using the TWFE design in equation 2. Table B4 reports these results. In contrast to the findings in Table B3, I find positive effects of the BS program expansion on men’s educational attainment. This likely captures broader district-level improvements in higher education supply that accompanied the BS expansion and benefited men as well, even though the program was not targeted at them.

Next, I replicate the spillover analysis of higher education expansion for women on men. Consistent with the underlying intuition, Table B5 shows no evidence of spillover effects of the expansion on men.

Alternate measures of outcome variable

The main analysis of the paper uses age-specific enrollment as the outcome variable. As a robustness check, I use years of schooling as the outcome variable in this section to examine the spillover effects. Table B8 reports the results when years of schooling are used as the outcome variable. The results show an increase in years of schooling at all pre-college levels of education as a result of exposure to the program.

Leave-One-District-Out Analyses

As I discussed above, given the non-random placement of the program across districts, the results could be driven by a single district. I address this concern by dropping one district at a time and re-estimating the analysis. The estimates, reported in Figures B4-B5, show that neither the direct nor the spillover effects are driven by any particular district.

Placebo Analyses

To further increase confidence in the results, I conduct a placebo exercise in this section. First, I randomly assign exposure to college among women in the sample and estimate the regression using this placebo treatment assignment. I repeat this procedure 1,000 times to obtain the distribution of placebo coefficients. The results from this exercise are reported in Figures B6-B8. These results further strengthen confidence in the main results.

Pre-college supply of schools

Another important concern regarding the spillover effects is that pre-college educational markets may be changing in response to increased access to higher education, rather than reflecting changes in parents' preferences to send their daughters to school. If this is the case, we would expect an increase in the supply of pre-college educational institutions in districts with greater access to higher education. To test this, I collect data on the supply of public primary and secondary schools from the Development Statistics provided by the KP Bureau of Statistics. In addition, some waves of the PSLM ask respondents about the distance to schools, which I also use to assess potential changes in pre-college educational markets. The results from these regressions are reported in Table B7. Column 1 of Table B7 shows that exposure to the program has no meaningful impact on distance to pre-college educational institutions. Similarly, Columns 2 and 3 show no statistically significant effect of program exposure on the supply of primary and secondary schools.²⁰

7 Conclusion

Existing evidence suggests that restrictive gender norms and safety concerns constrain women's mobility, thereby dampening higher-education enrollment in developing countries. At the same time, a growing body of work points to expansions in the local supply of tertiary institutions as a potentially powerful policy response. However, unlike investments at the primary and secondary levels, public spending on tertiary education is often politically contentious due to the distributional concerns.

In this paper, I evaluate a program where male-only colleges transitioned to coeducational

20. Unfortunately, there is no comparable data on the supply of private schools over time in KP. As shown in Table 5, enrollment has increased primarily in private schools; therefore, it cannot be ruled out that the private education market has responded to the expansion of higher education.

colleges, providing a cost-effective alternative to building new institutions for settings where single-sex educational institutions are still prevalent. Unfortunately, I am unable to exploit more credible variation generated by the staggered roll-out of the BS program across colleges and the subsequent introduction of coeducation within existing male-only colleges. Nevertheless, I use two-way fixed effects regressions to show that exposure to higher-education expansion increases female enrollment and the likelihood of completing a 16-year degree. Besides the direct effects, I document positive effects on marriage-market outcomes among exposed women. In addition, I find evidence of spillover effects on younger cohorts, suggesting that access to higher education shapes educational trajectories even before college-going age.

Taken together, this paper brings together two narratives. First, it is widely believed that coeducation may not be an appropriate policy response in conservative settings (Butt 2013; Karim 2024). Contrarily, I show that expanding access to higher education, even through the introduction of coeducation, led to increased female enrollment in KP province of Pakistan. Second, public spending in higher education is also believed to have low returns. In this regard, despite the absence of labor market returns for women, I show that the expansion of higher education has marriage market returns and spillover effects on younger cohorts.

Though the empirical evidence is weak, these findings still have two implications for policymakers. First, concerns that coeducation is inappropriate in conservative settings may be overstated, as I show that expanding higher education access through coeducation increased female enrollment even in KP province, widely regarded as a religiously and culturally conservative part of Pakistan. Second, assessments of public spending on higher education, just by focusing on labor market returns, may understate the overall benefits.

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Tables and Figures

Table 1: Effects on Educational, Marriage and Labor Market Outcomes

	Outcomes			
	Years of Schooling (1)	Graduated (2)	Married (3)	Employed (4)
Intensity	0.462* (0.269)	0.016** (0.006)	-0.048*** (0.013)	-0.008 (0.015)
Mean	3.687	0.052	0.886	0.102
Number of obs	50,469	50,469	50,486	50,486
R-squared	0.149	0.022	0.050	0.175
Birth-year FE	Y	Y	Y	Y
District FE	Y	Y	Y	Y
Survey-year FE	Y	Y	Y	Y

Notes: Each column reports coefficient estimates from separate regressions. The sample consists of women observed in the HIES 2018-19 and PSLM 2019-20. All regressions include birth-year, district, and survey-year fixed effects. Standard errors are reported in parentheses and are clustered at the district level. *p < 0.10; **p < 0.05; ***p < 0.01.

Table 2: Effects on Marriage Market Outcomes

	Outcomes						
	Age Diff. (1)	Educational Diff. (2)	Assortative Mating (3)	Hypergamy (4)	Homogamy (5)	Hypogamy (6)	Spouse Employed (7)
BSpc.girls	-0.144 (0.252)	-0.378* (0.211)	0.005 (0.004)	-0.040** (0.019)	0.042** (0.017)	-0.002 (0.009)	0.007 (0.010)
Observations	19,387	19,292	19,387	19,387	19,387	19,387	19,387
Mean Dep. Variable	3.383	3.831	0.013	0.562	0.352	0.086	0.862
R-squared	0.408	0.046	0.012	0.046	0.076	0.065	0.275
Birth-year FE	Y	Y	Y	Y	Y	Y	Y
District FE	Y	Y	Y	Y	Y	Y	Y
Survey-year FE	Y	Y	Y	Y	Y	Y	Y

Notes: Each column reports coefficient estimates from separate regressions. The sample consists of women observed in the HIES 2018–19 and PSLM 2019–20. All regressions include birth-year, district, and survey-year fixed effects. Standard errors are clustered at the district level and reported in parentheses. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table 3: Effects of BS Expansion on Educational Outcomes (Alternate Strategy)

	Outcomes	
	Currently Enrolled (1)	Years of Schooling (2)
Intensity	0.023*** (0.007)	0.363*** (0.128)
Mean	0.107	4.156
Number of obs	21,760	21,593
R-squared	0.070	0.146
Age FE	Y	Y
District FE	Y	Y
Survey-year FE	Y	Y

Notes: Each column reports coefficient estimates from separate regressions. The sample comprises women observed across all PSLM/HIES waves from 2005-06 to 2019-20. All regressions include age, district, and survey-year fixed effects. Standard errors are reported in parentheses and are clustered at the district level. *p < 0.10; **p < 0.05; ***p < 0.01.

Table 4: Spillover Effects of Expansion of BS Program on Pre-College Enrollment

	Currently Enrolled			
	Primary (1)	Secondary (2)	Matriculation (3)	Intermediate (4)
Intensity	0.012* (0.006)	0.016** (0.007)	0.022*** (0.007)	0.017*** (0.006)
Mean	0.591	0.507	0.359	0.226
Number of obs	56,244	22,062	13,566	13,239
R-squared	0.124	0.118	0.101	0.087
Age FE	Y	Y	Y	Y
District FE	Y	Y	Y	Y
Survey-year FE	Y	Y	Y	Y

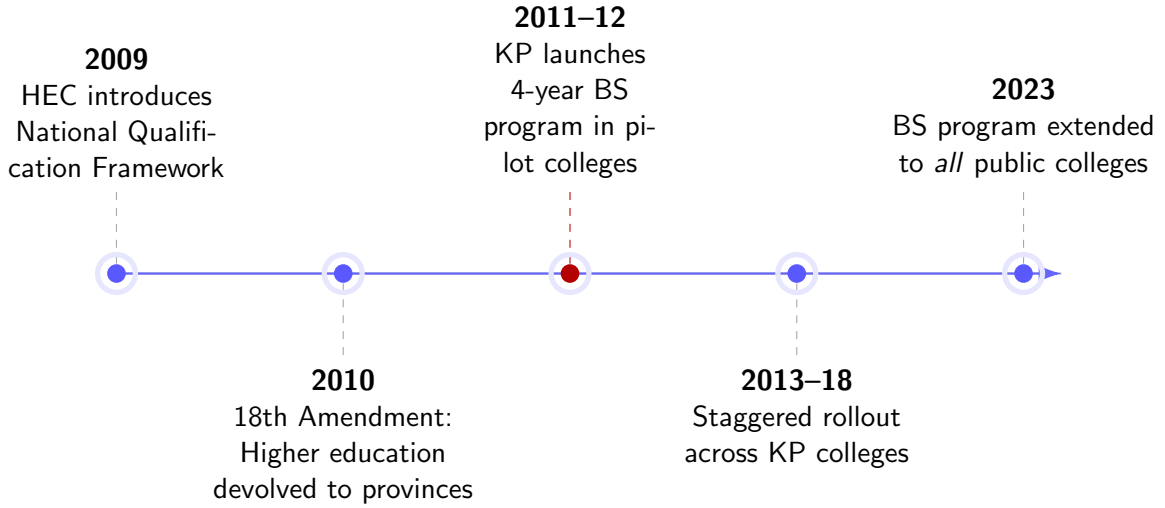
Notes: Each column reports coefficient estimates from separate regressions. The sample comprises women observed across all PSLM/HIES waves from 2005-06 to 2019-20. All regressions include age, district, and survey-year fixed effects. Standard errors are reported in parentheses and are clustered at the district level. *p < 0.10; **p < 0.05; ***p < 0.01.

Table 5: Dynamic Complementarities in Educational Investments

	Outcomes			
	Expenditures (1)	Remote (2)	Private (3)	Public (4)
Intensity	432.313*** (86.843)	-0.001*** (0.000)	0.015*** (0.003)	-0.001 (0.004)
Mean	5151.076	0.003	0.115	0.384
Number of obs	19,002	95,945	95,945	95,945
R-squared	0.228	0.009	0.078	0.083
Age FE	Y	Y	Y	Y
District FE	Y	Y	Y	Y
Survey-year FE	Y	Y	Y	Y

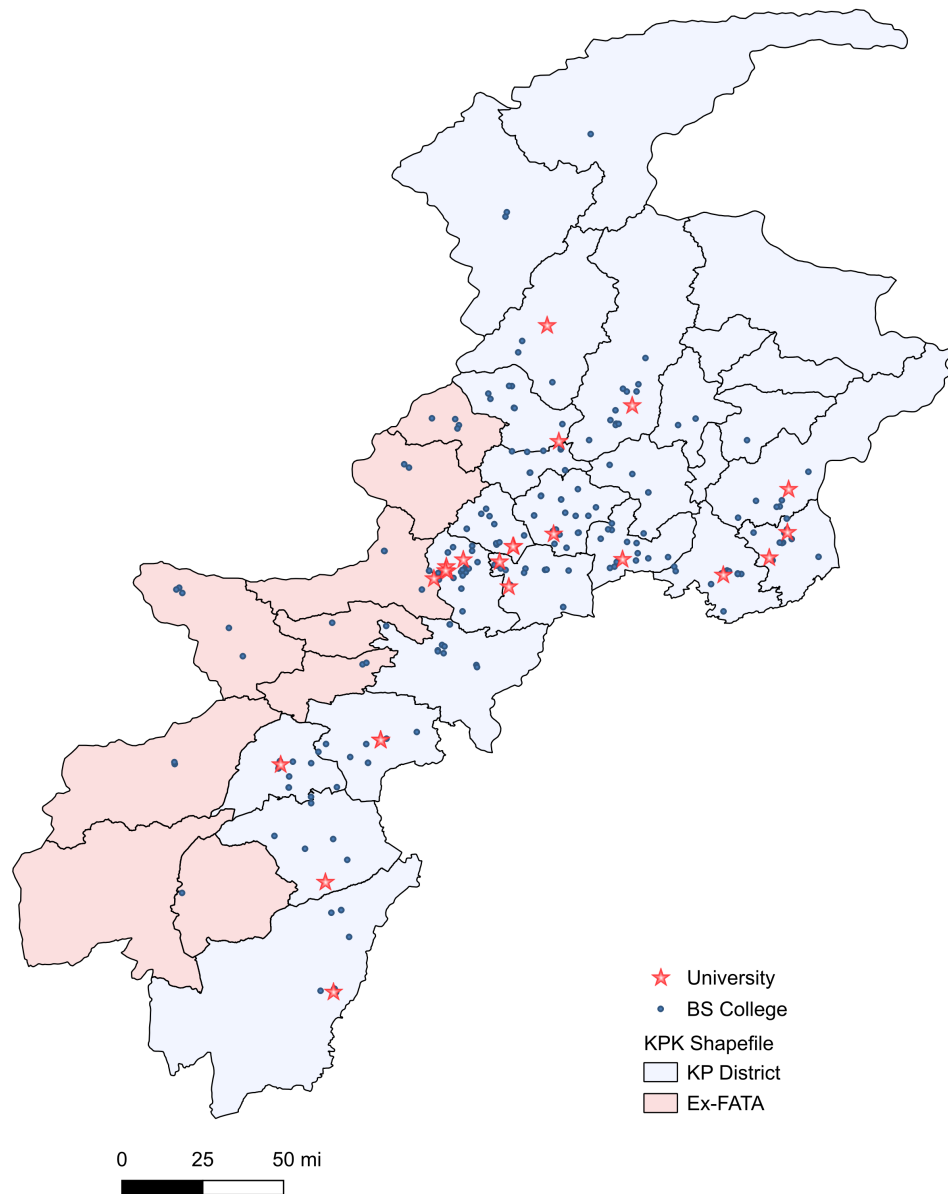
Notes: Each column reports coefficient estimates from separate regressions. The sample comprises women observed across all PSLM/HIES waves from 2005-06 to 2019-20. All regressions include age, district, and survey-year fixed effects. Standard errors are reported in parentheses and are clustered at the district level. *p < 0.10; **p < 0.05; ***p < 0.01.

Figure 1: Timeline of BS Program in Khyber Pakhtunkhwa province



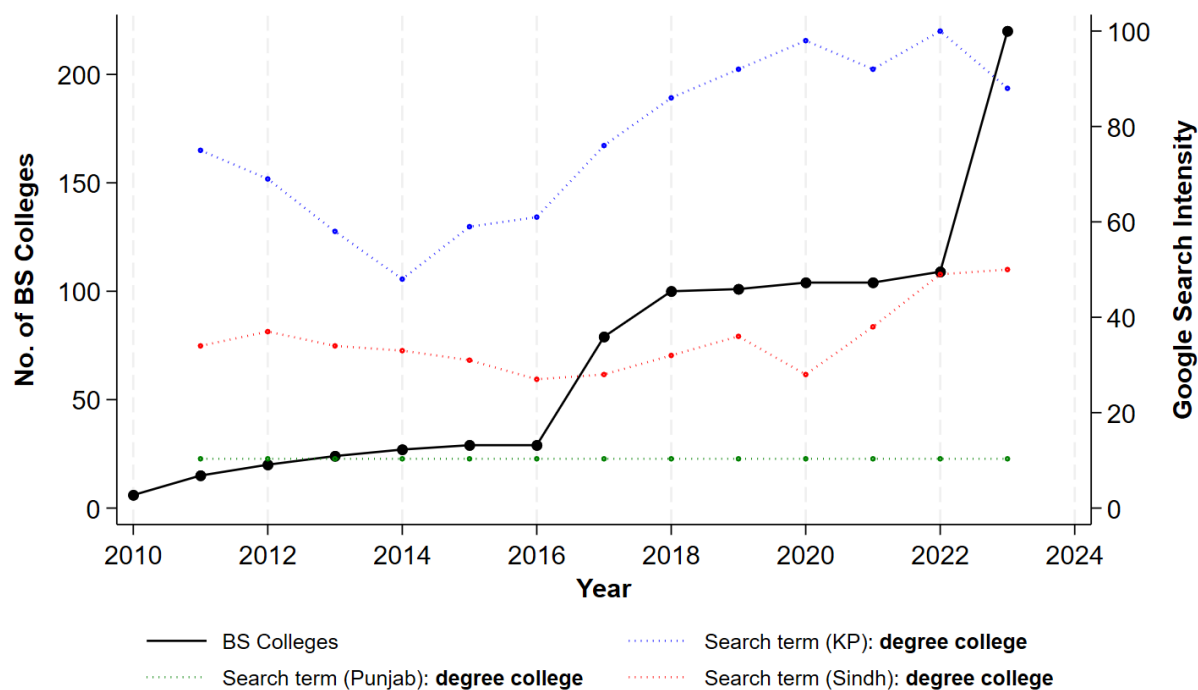
Notes: This figure summarizes key policy milestones related to the introduction and expansion of the four-year BS program in Khyber Pakhtunkhwa province.

Figure 2: Geographical distribution of Colleges and Universities in Khyber Pakhtunkhwa



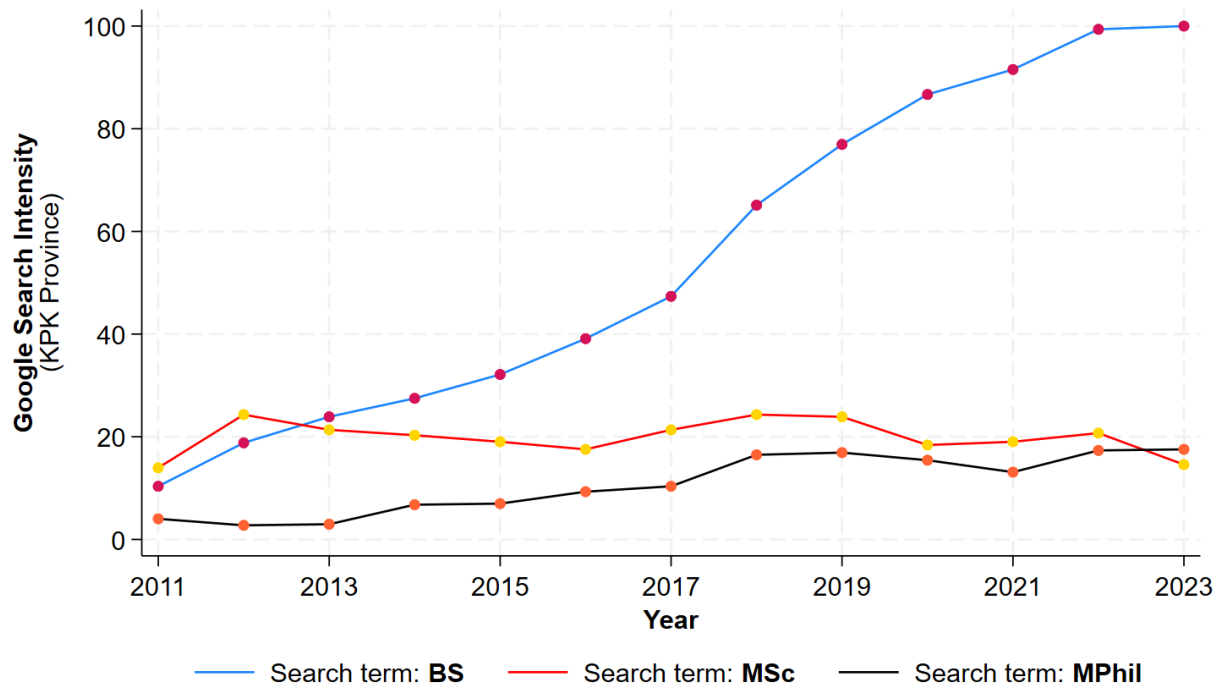
Notes: This figure maps the spatial distribution of higher-education institutions in Khyber Pakhtunkhwa. Blue dots denote BS colleges, while red stars indicate universities. District boundaries are shown in light gray, and districts formerly part of FATA are shaded in light red (not part of the analyses).

Figure 3: BS College Expansion and Google Search Intensity for Degree Colleges



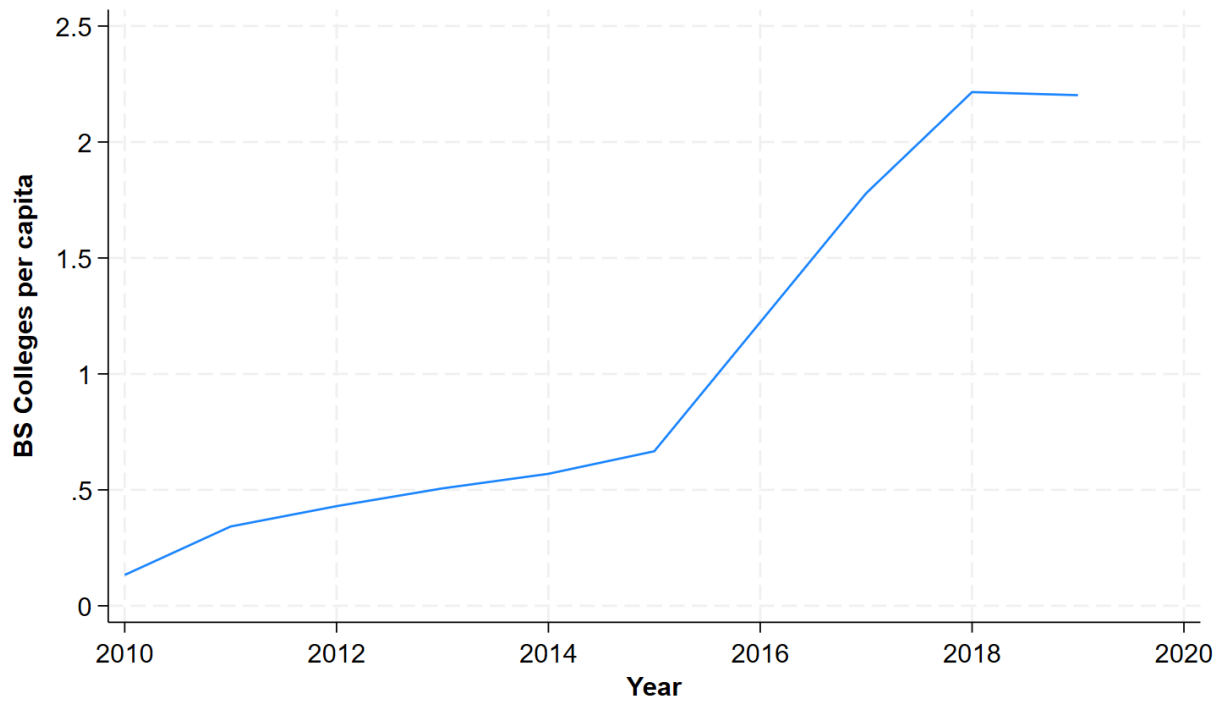
Notes: The figure plots the number of BS colleges in Khyber Pakhtunkhwa on left axis and Google search intensity for the term “degree college” in Khyber Pakhtunkhwa, Punjab, and Sindh on right axis. Google search intensity is aggregated at year-level.

Figure 4: Google Search Intensity for BS, MSc, and MPhil in Khyber Pakhtunkhwa



Notes: The figure plots annual Google search intensity for the terms “BS”, “MSc”, and “MPhil” in Khyber Pakhtunkhwa. Google search intensity is aggregated at the year-level.

Figure 5: BS Colleges per 100,000 Females in Khyber Pakhtunkhwa over Time



Notes: The figure shows BS colleges per 100,000 college-going females in Khyber Pakhtunkhwa over time.

Online Appendix

**“When Girls Get a Chance: Direct and Spillover
Effects of Female Access to Higher Education in
Pakistan”**

Appendix A: Additional Figures and Tables

Table B1: Descriptive Statistics: Cohort Analyses Sample

	N	Mean	SD	Min	Max
<i>Panel A: Direct Effects</i>					
Years Schooling	50,469	3.687	4.782	0	18
Master's Degree	50,469	0.052	0.175	0	1
Married	50,486	0.898	0.303	0	1
Worked	50,486	0.102	0.303	0	1
<i>Panel B: Spousal Analyses</i>					
Age Difference	19,387	3.383	8.318	-30	73
Education Difference	19,292	3.831	5.245	-16	18
Assortative Matching (Masters)	19,387	0.013	0.113	0	1
Hypergamy	19,387	0.562	0.496	0	1
Homogamy	19,387	0.352	0.478	0	1
Hypogamy	19,387	0.086	0.281	0	1
Spouse Worked	19,387	0.862	0.345	0	1

Table B2: Descriptive Statistics: Indirect Effects Sample

	N	Mean	SD	Min	Max
<i>Panel A: Direct Effects</i>					
Current Enrolled	21,761	0.107	0.310	0	1
Years Schooling	21,593	4.156	5.311	0	22
<i>Panel B: Spillover Effects</i>					
Enrolled Primary	56,244	0.591	0.492	0	1
Enrolled Secondary	22,063	0.507	0.500	0	1
Enrolled Matriculation	13,567	0.359	0.480	0	1
Enrolled Intermediate	13,240	0.226	0.418	0	1
<i>Panel C: BS Colleges Data</i>					
Intensity	230	0.983	1.225	0	6.123

Table B3: Effects on Educational, Marriage and Labor Market Outcomes (Male Sample)

	Outcomes			
	Years of Schooling (1)	Graduated (2)	Married (3)	Employed (4)
Intensity	0.111 (0.163)	0.006 (0.007)	-0.069*** (0.021)	-0.002 (0.014)
Mean	6.985	0.080	0.784	0.871
Number of obs	36,666	36,666	36,944	36,944
R-squared	0.084	0.022	0.240	0.062
Birth-year FE	Y	Y	Y	Y
District FE	Y	Y	Y	Y
Survey-year FE	Y	Y	Y	Y

Notes: Each column reports coefficient estimates from separate regressions. The sample consists of males observed in the HIES 2018-19 and PSLM 2019-20. All regressions include birth-year, district, and survey-year fixed effects. Standard errors are reported in parentheses and are clustered at the district level. *p < 0.10; **p < 0.05; ***p < 0.01.

Table B4: Effects of BS Expansion on Educational Outcomes (Male Sample)

	Outcomes	
	Currently Enrolled (1)	Years of Schooling (2)
Intensity	0.011** (0.005)	0.190** (0.069)
Mean	0.282	8.205
Number of obs	20,016	19,414
R-squared	0.049	0.074
Age FE	Y	Y
District FE	Y	Y
Survey-year FE	Y	Y

Notes: Each column reports coefficient estimates from separate regressions. The sample comprises men observed across all PSLM/HIES waves from 2005-06 to 2019-20. All regressions include age, district, and survey-year fixed effects. Standard errors are reported in parentheses and are clustered at the district level. *p < 0.10; **p < 0.05; ***p < 0.01.

Table B5: Spillover Effects of Expansion of BS Program on Pre-College Enrollment (Male Sample)

	Currently Enrolled			
	Primary (1)	Secondary (2)	Matriculation (3)	Intermediate (4)
Intensity	-0.000 (0.004)	-0.005 (0.004)	0.006 (0.006)	-0.013** (0.006)
Mean	0.692	0.695	0.550	0.375
Number of obs	63,483	24,831	15,404	14,235
R-squared	0.162	0.048	0.051	0.055
Age FE	Y	Y	Y	Y
District FE	Y	Y	Y	Y
Survey-year FE	Y	Y	Y	Y

Notes: Each column reports coefficient estimates from separate regressions. The sample comprises men observed across all PSLM and HIES waves from 2005-06 to 2019-20. All regressions include age, district, and survey-year fixed effects. Standard errors are reported in parentheses and are clustered at the district level. *p < 0.10; **p < 0.05; ***p < 0.01.

Table B6: Dynamic Complementarities in Educational Investments (Male Sample)

	Outcomes			
	Expenditures (1)	Remote (2)	Private (3)	Public (4)
Intensity	621.842*** (96.926)	-0.000 (0.000)	0.016*** (0.004)	-0.019*** (0.005)
Mean	7028.289	0.003	0.199	0.557
Number of obs	24,733	108,114	108,114	108,114
R-squared	0.206	0.008	0.070	0.046
Age FE	Y	Y	Y	Y
District FE	Y	Y	Y	Y
Survey-year FE	Y	Y	Y	Y

Notes: Each column reports coefficient estimates from separate regressions. The sample comprises men observed across all Pakistan Social and Living Standards Measurement (PSLM) and Household Integrated Economic Survey (HIES) waves from 2005–06 to 2019–20. All regressions include age, district, and survey-year fixed effects. Standard errors are reported in parentheses and are clustered at the district level. *p < 0.10; **p < 0.05; ***p < 0.01.

Table B7: Effects on School Availability and Distance to Schools

	Outcomes		
	Distance (1)	Primary (2)	Secondary (3)
Intensity	0.143 (0.129)	-8.214 (12.518)	1.473 (1.516)
Mean	0.099	330.712	42.529
Number of obs	10,297	240	240
R-squared	0.088	0.990	0.987
Age FE	Y	N/A	N/A
District FE	Y	Y	Y
Survey-year FE	Y	N/A	N/A

Notes: Each column reports coefficient estimates from separate regressions. Standard errors are reported in parentheses and are clustered at the district level. *p < 0.10; **p < 0.05; ***p < 0.01.

Table B8: Impact of BS College Expansion Intensity on Years of Schooling by Level

	Years of Schooling			
	Primary (1)	Secondary (2)	Matriculation (3)	Intermediate (4)
Intensity	0.040** (0.014)	0.132** (0.052)	0.248*** (0.083)	0.208** (0.090)
Number of obs	56,077	21,920	13,448	13,095
R-squared	0.367	0.145	0.143	0.152
Age FE	Y	Y	Y	Y
District FE	Y	Y	Y	Y
Survey-year FE	Y	Y	Y	Y

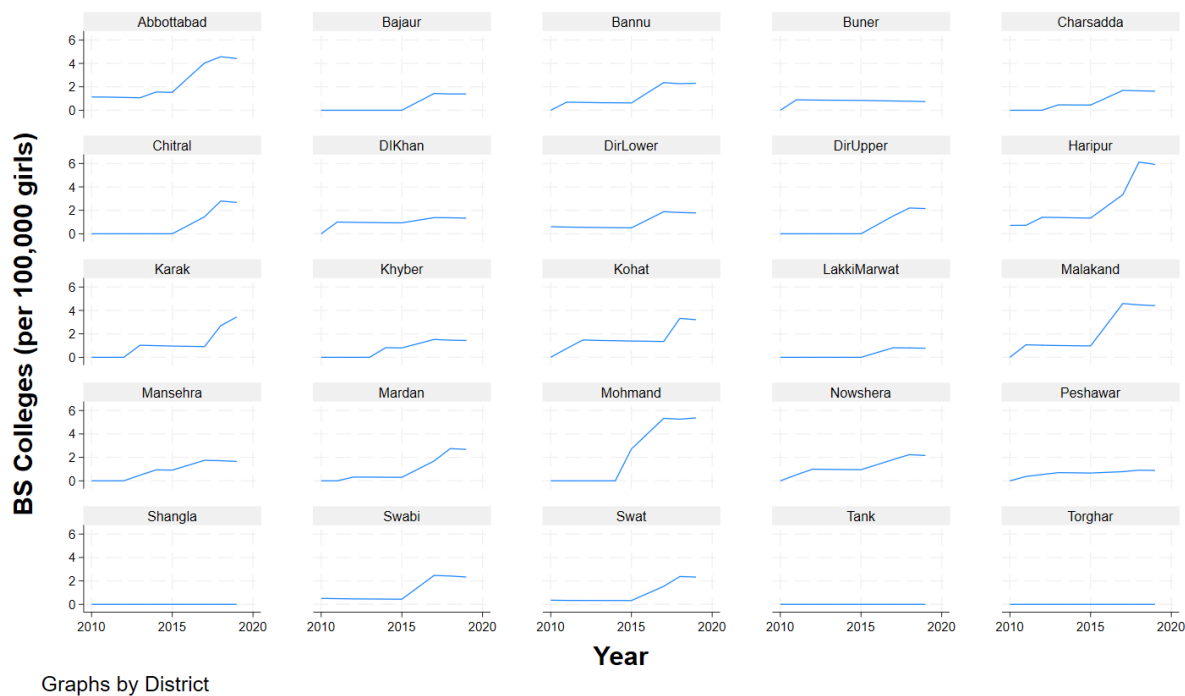
Notes: Each column reports coefficient estimates from separate regressions of years of schooling accumulated at the specified educational level on the intensity of BS college expansion. The sample consists of school-age females observed in the household survey data. All regressions include age, district, and survey-year fixed effects. Standard errors are reported in parentheses. *p < 0.10; **p < 0.05; ***p < 0.01.

Table B9: Timing of BS College Expansion Across Districts

Years	Untreated Districts	Treated Districts	Total
2010	21	5	26
2011	14	12	26
2012	13	13	26
2013	10	16	26
2014	9	17	26
2015	7	19	26
2017	3	23	26
2018	3	23	26
2019	3	23	26
2020	3	23	26
2022	3	23	26
2023	0	26	26

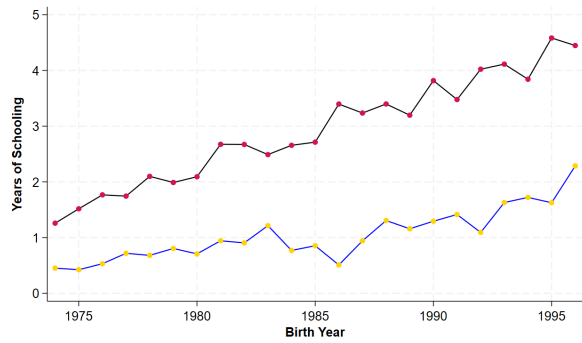
Notes: This table reports the number of districts with and without a BS college in each year. Treated districts are those that have introduced at least one BS college by the corresponding year, while untreated districts have not yet been exposed.

Figure B1: District-level Expansion of BS Colleges per 100,000 Girls

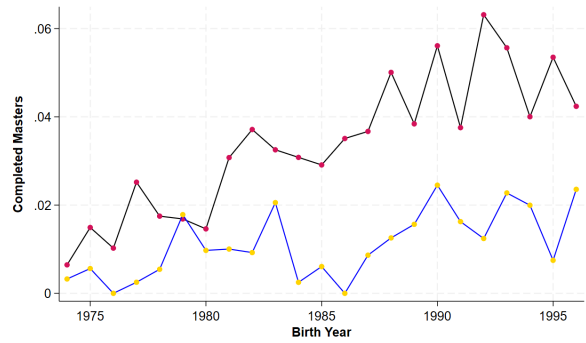


Notes: This figure plots the number of BS colleges per 100,000 girls over time for each district. Each panel corresponds to a district and shows the evolution of exposure to BS college expansion between 2010 and 2020.

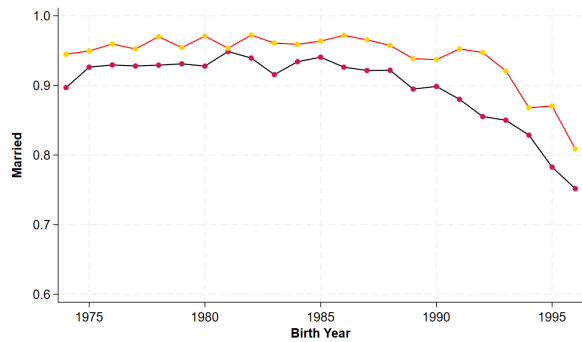
Figure B2: Parallel Trend (Cohort Analyses)



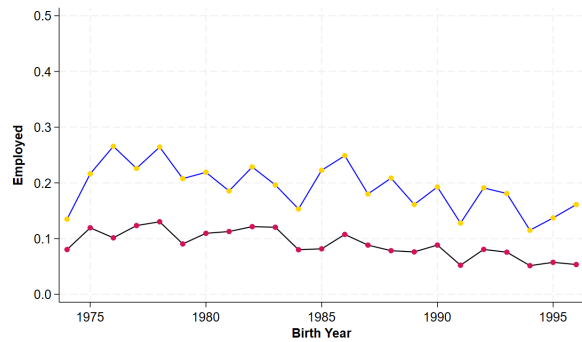
(a) Years of Schooling



(b) Masters

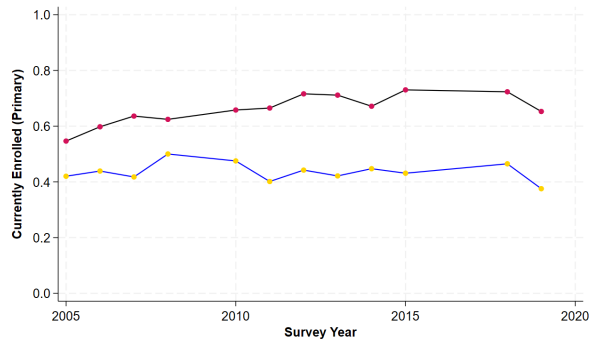


(c) Marriage

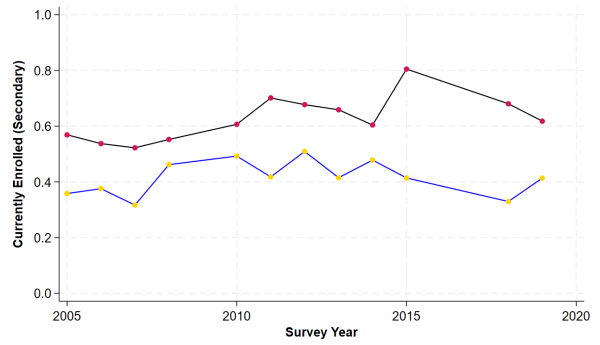


(d) Labor Force Participation

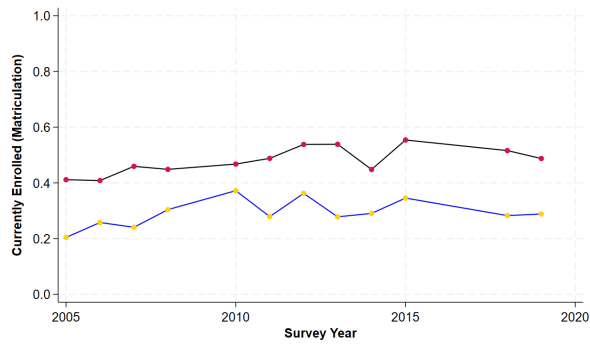
Figure B3: Parallel Trend (Spillover Effects)



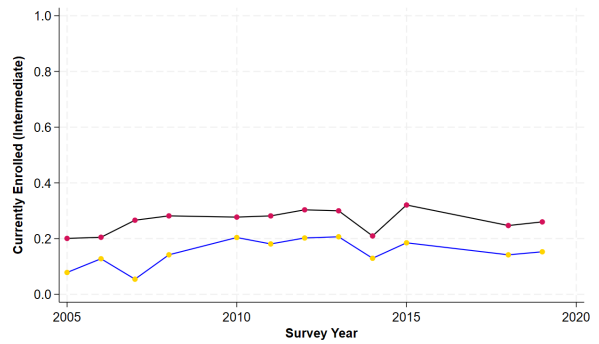
(a) Primary



(b) Secondary

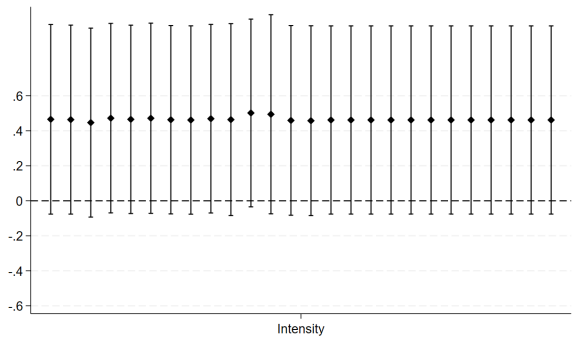


(c) Matriculation

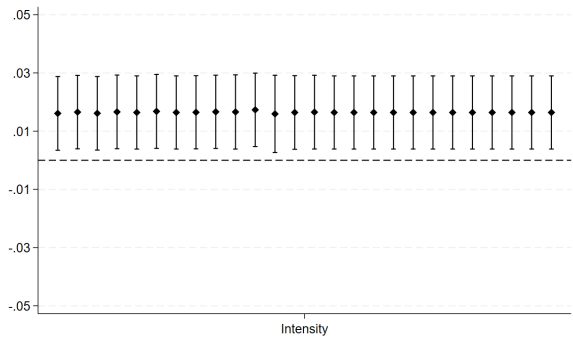


(d) Intermediate

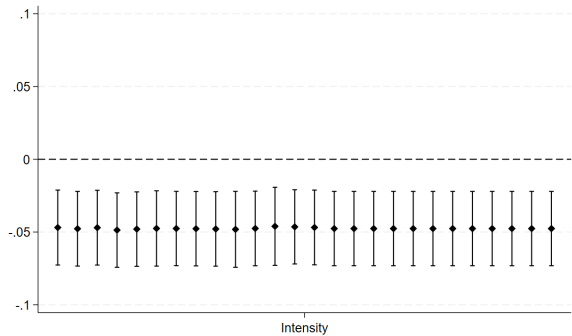
Figure B4: Robustness Analyses (Cohort Analysis)



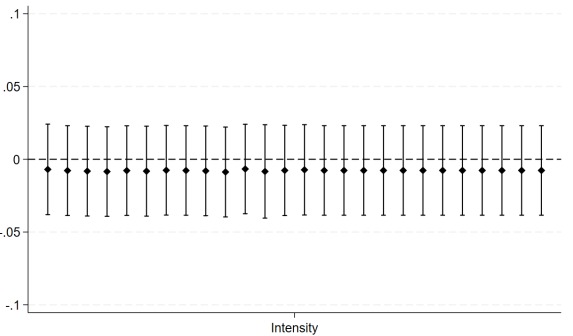
(a) Years of Schooling



(b) Masters

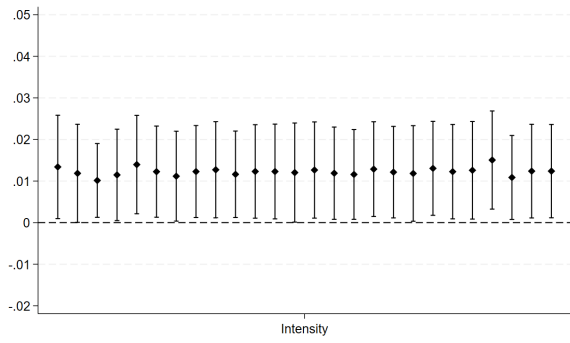


(c) Marriage

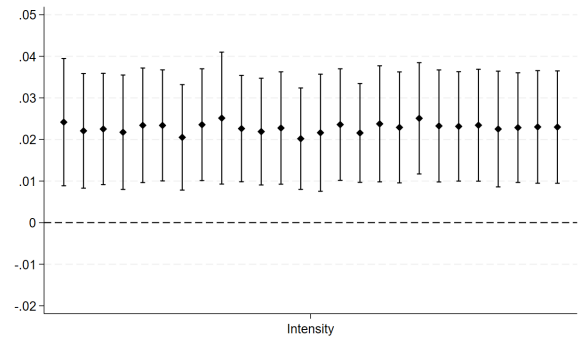


(d) LFP

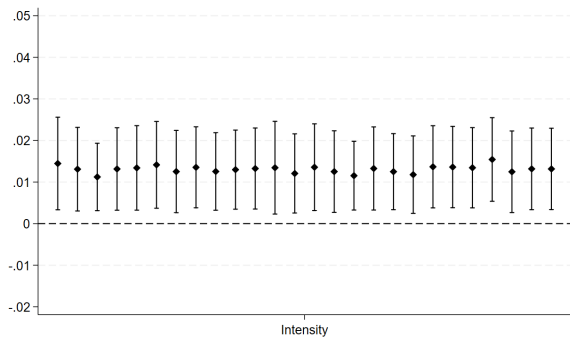
Figure B5: Robustness Analyses (Spillover Effects)



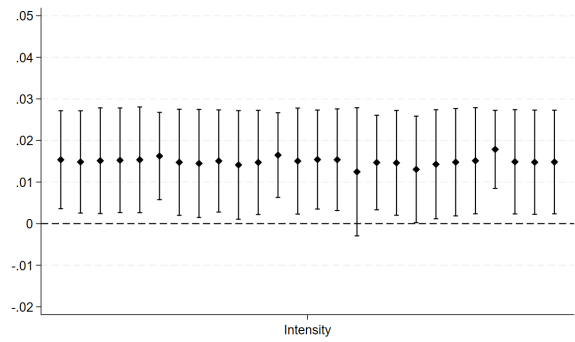
(a) Primary



(b) Secondary

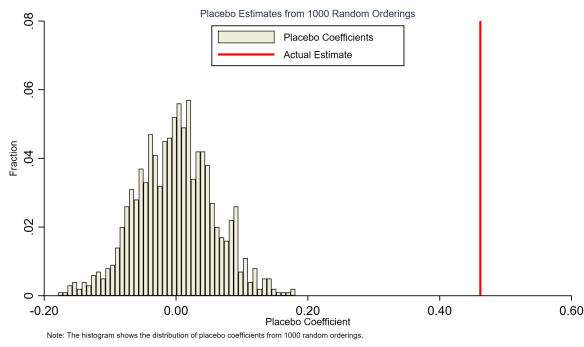


(c) Matriculation

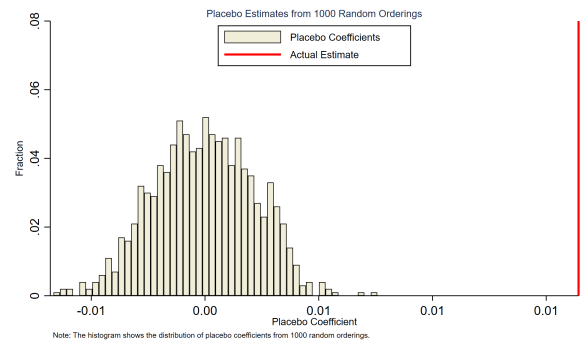


(d) Intermediate

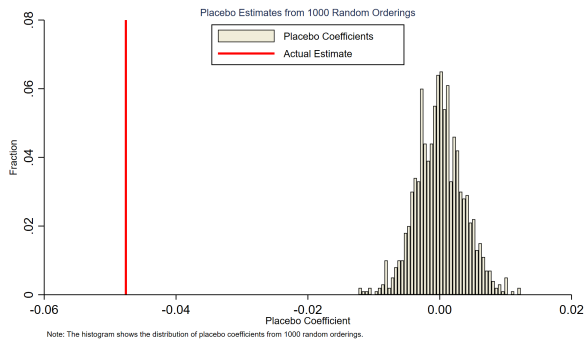
Figure B6: Placebo Analyses (Cohort Analysis)



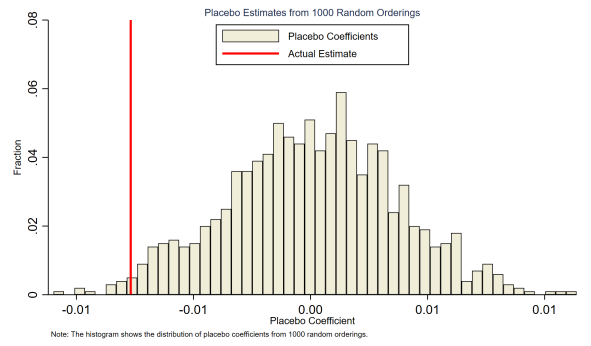
(a) Years of Schooling



(b) Masters

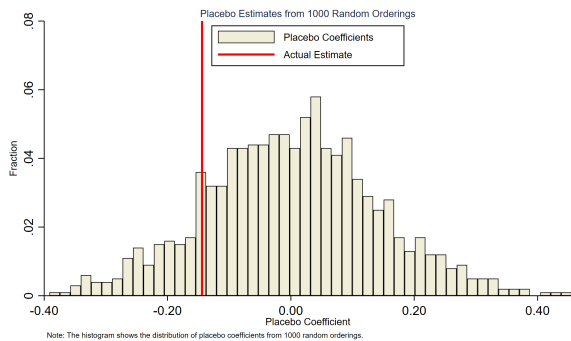


(c) Marriage

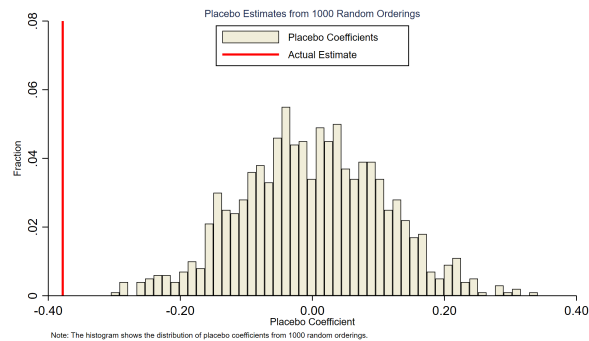


(d) Labor Force Participation

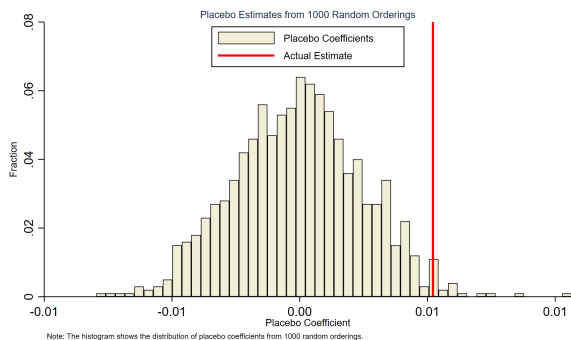
Figure B7: Placebo Analyses (Marriage Market Outcomes)



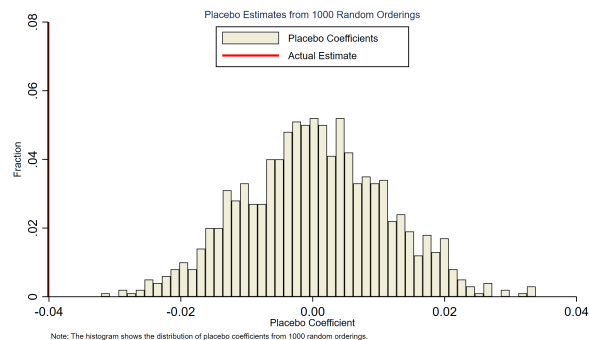
(a) Spousal Age Difference



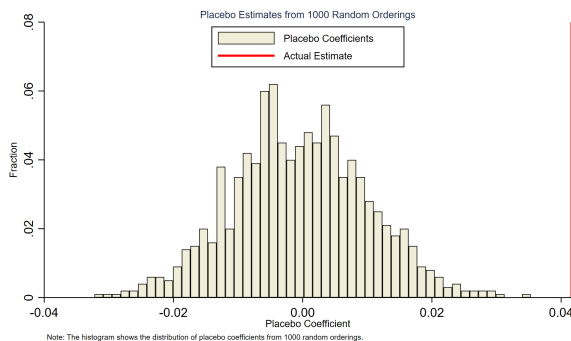
(b) Spousal Education Difference



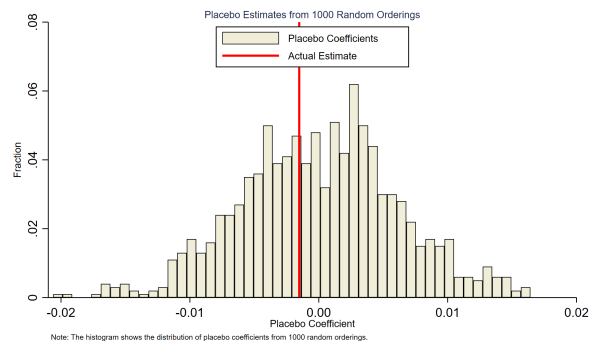
(c) Assortative Mating



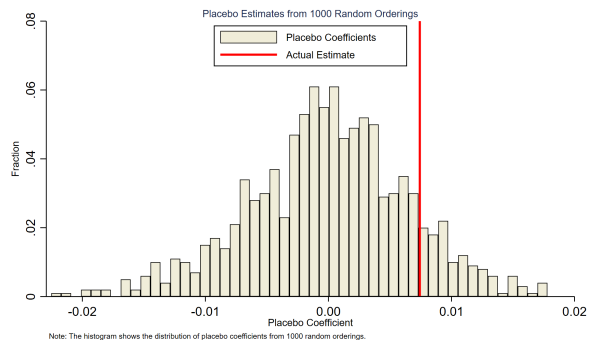
(d) Hypergamy



(e) Homogamy

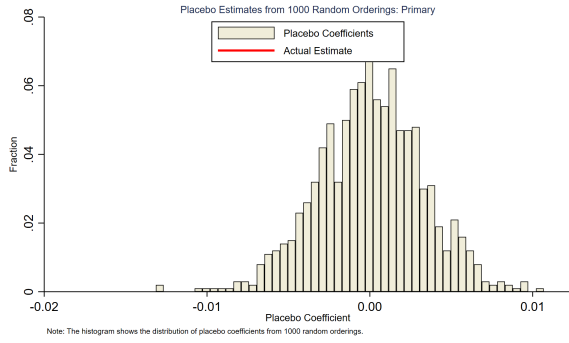


(f) Hypogamy

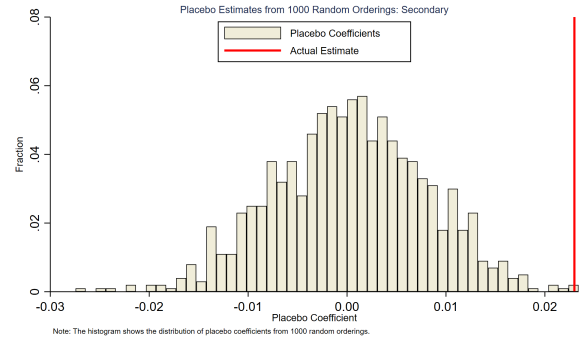


(g) Spouse Worked

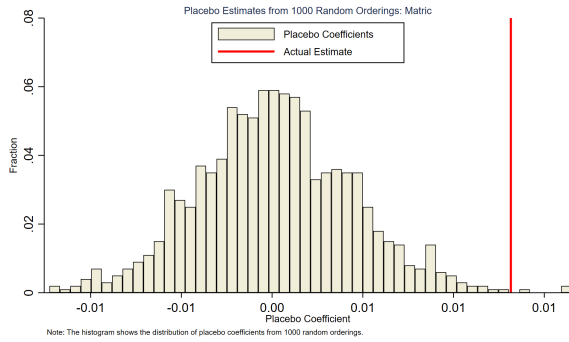
Figure B8: Placebo Analyses (Spillover Effects)



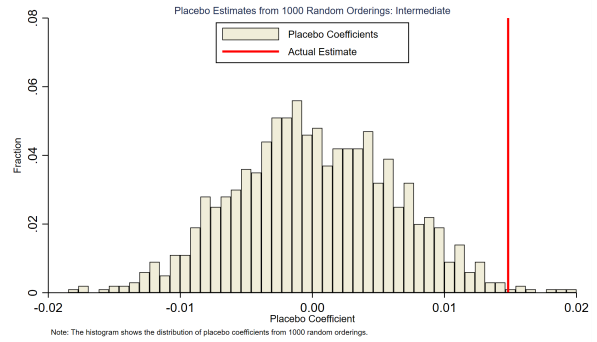
(a) Primary



(b) Secondary



(c) Matriculation



(d) Intermediate